THE INCLUSION OF THE BIG AFFINE H SITE INTO THE BIG H SITE IS COCONTINUOUS AND INDUCES AN EQUIVALENCE OF TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EU4 (sources/stacks-flat/pinned-original-source.tex line 10572).

Allowed standard prerequisites (editorial).

- Special cocontinuous functors and the five-part criterion for an equivalence of topoi. Sites, Definition labelled sites-definition-special-cocontinuous-functor (Stacks tag 03CG) and Lemma labelled sites-lemma-equivalence (Stacks tag 03A0), invoked at proof lines 10582-10585. Supply the list of five conditions that the proof checks one by one.
- Every h covering of an affine scheme is refined by a standard h covering. Same chapter, Lemma labelled lemma-approximate-h-cover (source line 10303), invoked at proof lines 10588-10590. Supplies condition (1), the only step that uses the h topology in an essential way.

Source reference. Original definition 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10583 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10585 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ETU: sources/stacks-flat/pinned-original-source.tex, source lines 10303–10303. Same-chapter pointer into the author environment lemma

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at source line 10303; invoked at proof line 10590 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a scheme. Let Sch_h be a big h site containing S. The functor $(Aff/S)_h \to (Sch/S)_h$ is cocontinuous and induces an equivalence of topoi from $Sh((Aff/S)_h)$ to $Sh((Sch/S)_h)$.*

Proof. The notion of a special cocontinuous functor is introduced in Sites, Definition 03CG. Thus we have to verify assumptions (1) – (5) of Sites, Lemma 03A0. Denote the inclusion functor $u : (Aff/S)_h \to (Sch/S)_h$. Being cocontinuous follows because any h covering of T/S, T affine, can be refined by a standard h covering for example by Lemma 0ETU. Hence (1) holds. We see u is continuous simply because a standard h covering is a h covering. Hence (2) holds. Parts (3) and (4) follow immediately from the fact that u is fully faithful. And finally condition (5) follows from the fact that every scheme has an affine open covering (which is a h covering). □